THE PUSHFORWARD ALONG A CLOSED IMMERSION IS FULLY FAITHFUL, WITH ESSENTIAL IMAGE DESCRIBED BY SUPPORT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument for both parts of the statement and for the final assertion about the inverse image.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter etale-cohomology.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- Closed immersion and the comparison of the etale sites of X and of the complement. Same chapter, the open immersion of the complement of Z and the resulting picture in which a sheaf on X_etale is determined by its restriction to Z and to the complement. Supplies the two functors compared and the description of the essential image.
- Adjunction between the small inverse image and pushforward along a closed immersion. Same chapter, the adjunction i_{small}^{-1} -| i_{small,*} and the fact that the inverse image is a left inverse of the pushforward. Supplies the fully faithfulness and the last assertion of the statement.

Source reference. Original lemma 04FX: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 6644–6644. Navigation pointer invoked at proof line 6734; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original theorem 03PU: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 3546–3546. Navigation pointer invoked at proof line 6736; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Date: stacks-project:etale-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 04D6: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 6738–6738. Navigation pointer invoked at proof line 6738; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 03Q1: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 4649–4649. Navigation pointer invoked at proof line 6748; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Proposition 0.1. *Let $i : Z \to X$ be a closed immersion of schemes.*

(1) *The functor*
$$i_{small,*} : Sh(Z_{étale}) \longrightarrow Sh(X_{étale})$$
is fully faithful and its essential image is those sheaves of sets $\mathcal{F}$ on $X_{étale}$ whose restriction to $X \setminus Z$ is isomorphic to $*$, and

(2) *the functor*
$$i_{small,*} : Ab(Z_{étale}) \longrightarrow Ab(X_{étale})$$
is fully faithful and its essential image is those abelian sheaves on $X_{étale}$ whose support is contained in Z.

In both cases i_{small}^{-1} is a left inverse to the functor $i_{small,*}$.

Proof. Let's discuss the case of sheaves of sets. For any sheaf $\mathcal{G}$ on Z the morphism $i_{small}^{-1} i_{small,*}\mathcal{G} \to \mathcal{G}$ is an isomorphism by Lemma 04FX (and Theorem 03PU). This implies formally that $i_{small,*}$ is fully faithful, see Sites, Lemma 04D6. It is clear that $i_{small,*}\mathcal{G}|_{U_{étale}} \cong *$ where $U = X \setminus Z$. Conversely, suppose that $\mathcal{F}$ is a sheaf of sets on X such that $\mathcal{F}|_{U_{étale}} \cong *$. Consider the adjunction mapping
$$\mathcal{F} \longrightarrow i_{small,*} i_{small}^{-1} \mathcal{F}$$
Combining Lemmas 04FX and 03Q1 we see that it is an isomorphism. This finishes the proof of (1). The proof of (2) is identical. □